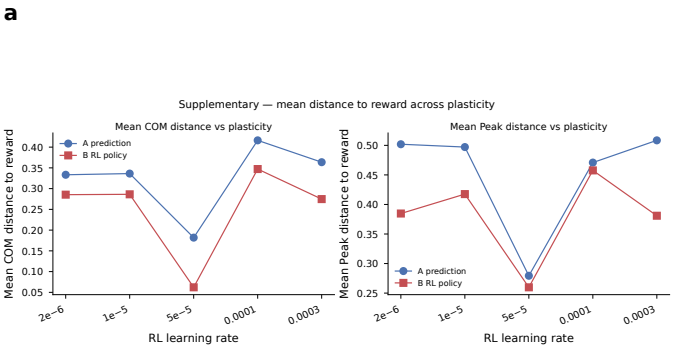
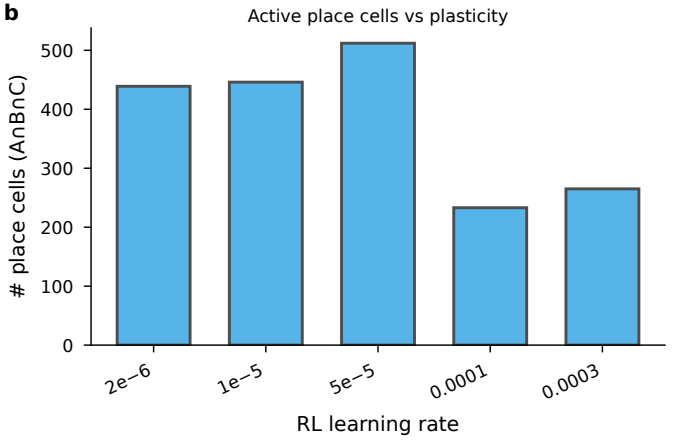
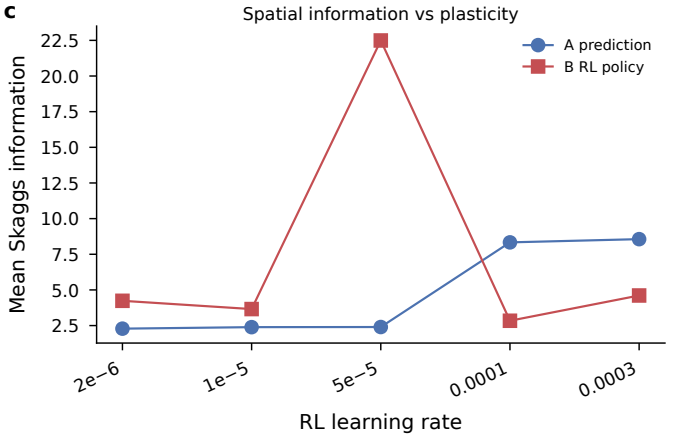
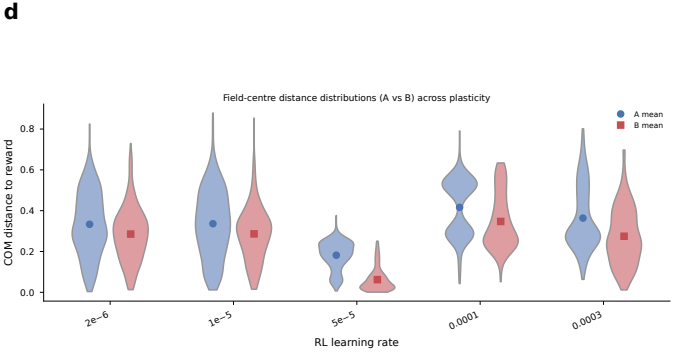

a
Supplementary — mean distance to reward across plasticity
Mean COM distance vs plasticity
Mean COM distance to reward
A prediction
B RL policy
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
2e−6
1e−5
5e−5
0.0001
0.0003
RL learning rate
Mean Peak distance vs plasticity
Mean Peak distance to reward
A prediction
B RL policy
0.50
0.45
0.40
0.35
0.30
0.25
2e−6
1e−5
5e−5
0.0001
0.0003
RL learning rate


b
Active place cells vs plasticity
place cells (AnBnC)
500
400
300
200
100
0
2e−6
1e−5
5e−5
0.0001
0.0003
RL learning rate


c
Spatial information vs plasticity
Mean Skaggs information
A prediction
B RL policy
22.5
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
2e−6
1e−5
5e−5
0.0001
0.0003
RL learning rate


d
Field-centre distance distributions (A vs B) across plasticity
COM distance to reward
A mean
B mean
0.8
0.6
0.4
0.2
0.0
2e−6
1e−5
5e−5
0.0001
0.0003
RL learning rate